BUSINESS FLOWS

Business Flows Documentation

This document outlines all major business processes and workflows in the Oro Ticket App system.

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1. User Authentication Flow

Primary Flow: First-Time User Setup

Detailed Steps:

1. $1
User downloads and installs Oro Ticket App
App detects first installation via Hive storage check

2. $1
App shows reset password screen for initial setup
User creates strong password
Password stored securely in device storage

3. $1
User enters email and password
App validates credentials against API
JWT token received and stored securely

4. $1
Fetch user profile data from server
Load company information and permissions
Sync critical data (vehicles, terminals, commission rules)

5. $1
User redirected to main dashboard
App loads user's assigned vehicles and routes

Alternative Flow: Returning User

Session Management:
$1: Automatic token expiry check
$1: Cached credentials for offline access
$1: Unsynced data prevents logout

2. Ticket Booking and Issuance Flow

Primary Flow: Standard Ticket Issuance

Detailed Steps:

1. $1
User selects from assigned company vehicles
App validates vehicle availability and status
Loads vehicle-specific tariffs and routes

2. $1
Select departure terminal (auto-assigned based on user profile)
Choose arrival terminal from available destinations
System validates route availability

3. $1

4. $1
System generates unique ticket data
Formats ticket with Ethiopian date/time
Includes QR code for verification

5. $1
Connects to Bluetooth thermal printer
Prints formatted ticket with company branding
Multiple copies based on seat count

6. $1
Stores trip details in local Hive database
Marks trip as pending synchronization
Updates daily service charge totals

Edge Cases:
$1: All steps work without internet
$1: Manual ticket issuance with digital record
$1: System prevents invalid terminal combinations
$1: Real-time vehicle status checking

3. Trip Management Flow

Daily Trip Operations

Detailed Steps:

1. $1
Driver checks assigned vehicle
Reviews scheduled routes and timings
Verifies vehicle condition and fuel

2. $1
Passenger boarding and ticket collection
Route navigation and timing adherence
Emergency handling procedures

3. $1
Record actual trip metrics (distance, time, passengers)
Update vehicle status and maintenance needs
Calculate actual revenue vs. projected

4. $1
Upload trip data to central server
Receive updated schedules and assignments
Sync with dispatch system

Trip Status Management:
$1: Planned trips in system
$1: Currently active trips
$1: Successfully finished trips
$1: Cancelled or aborted trips

4. Data Synchronization Flow

Offline-First Synchronization Strategy

Synchronization Types:

1. $1 (When Online)
Automatic background synchronization
Immediate data upload/download
Real-time conflict detection

2. $1 (Manual Trigger)
User-initiated synchronization
Bulk data transfer
Progress indicators and error handling

3. $1 (Automated)
Regular interval synchronization
Low-priority background processing
Bandwidth-aware scheduling

Data Synchronization Categories:
$1: Vehicles, terminals, commission rules
$1: Trips, service charges, tickets
$1: User profiles, company settings

Conflict Resolution Strategies:
$1: Server data takes precedence
$1: Local changes override server
$1: User intervention required
$1: Intelligent data merging

5. Reporting and Analytics Flow

Report Generation Process

Report Types:

1. $1
Daily trip summaries
Route performance analysis
Revenue reports by vehicle/route

2. $1
Daily service charge summaries
Employee performance tracking
Commission calculations

3. $1
Vehicle usage statistics
Maintenance scheduling
Fuel consumption analysis

4. $1
Revenue analysis
Commission payouts
Profit/loss statements

Report Generation Steps:

1. $1
Collect data from local storage
Apply date/time filters
Group by relevant categories

2. $1
Calculate totals and averages
Apply business rules
Format for presentation

3. $1
Create formatted PDF documents
Include charts and tables
Add company branding

4. $1
Save to device storage
Email distribution
Cloud storage upload

6. Fleet Management Flow

Vehicle Lifecycle Management

Detailed Steps:

1. $1
Enter vehicle details (plate number, capacity, type)
Associate with transport company
Configure available routes and terminals

2. $1
Assign departure terminals
Configure arrival destinations
Set tariff rates per route

3. $1
Track daily assignments
Monitor vehicle status
Record maintenance schedules

4. $1
Monitor fuel efficiency
Track revenue generation
Analyze route profitability

5. $1
Schedule regular maintenance
Track repair history
Monitor vehicle health

7. Service Charge Management Flow

Commission Calculation and Tracking

Service Charge Components:

1. $1

2. $1
Per trip charges
Daily totals per employee
Weekly/monthly summaries

3. $1
Real-time charge updates
Historical data preservation
Audit trail maintenance

4. $1
Commission calculation
Payment scheduling
Financial reporting

8. Commission Management Flow

Commission Rule Administration

Commission Rule Types:

1. $1: Fixed percentage of ticket price
2. $1: Different rates based on volume/sales
3. $1: Different rates per route or terminal
4. $1: Seasonal or time-specific rates

Rule Management Process:

1. $1
Define commission parameters
Set applicability criteria
Configure calculation methods

2. $1
Link rules to user roles
Set effective dates
Configure overrides

3. $1
Automatic rule application
Real-time commission calculation
Transparent fee disclosure

4. $1
Performance tracking
Rule effectiveness analysis
Periodic adjustments

Business Rules and Constraints

Ticket Issuance Rules
$1: Cannot exceed registered seat capacity
$1: Only predefined routes allowed
$1: Tickets valid for scheduled trip times
$1: Unique ticket numbers per trip

Financial Rules
$1: Maximum commission rates enforced
$1: Automatic tax application where required
$1: Ethiopian Birr (ETB) as base currency
$1: Consistent rounding for financial calculations

Operational Rules
$1: Automatic logout after inactivity
$1: Configurable data retention periods
$1: Regular data backup procedures
$1: Complete audit trail for all transactions

Error Handling and Recovery

Common Error Scenarios

1. $1
Automatic retry mechanisms
Offline mode activation
Data queuing for later sync

2. $1
Alternative ticket formats
Digital ticket options
Error logging and reporting

3. $1
Conflict detection algorithms
User notification and resolution
Automatic merge strategies

4. $1
Token refresh mechanisms
Offline authentication fallback
Secure credential recovery

Performance Considerations

System Performance Requirements
$1: < 2 seconds for UI interactions
$1: < 30 seconds for data synchronization
$1: Optimized local data storage
$1: Minimal impact on device battery

Scalability Considerations
$1: Support for large vehicle fleets
$1: Multi-user operation support
$1: Bandwidth-optimized synchronization
$1: Automatic data cleanup and archiving

$1: 1.0
$1: November 2024
$1: Oromia Transport Agency Operations Department